

When Aerodynamic States Become Latent: Reserve-Risk Screening under Degraded Wind-Turbine SCADA Telemetry

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Abstract—Operating reserve screening near the demarcation boundary between Maximum Power Point Tracking (MPPT, Region 2) and active blade-pitch regulation (Region 3) is governed by acute asymmetric shortfall risk: generation over-forecasts incur severe balancing penalties relative to surplus energy headroom. Deterministic aerodynamic power curves presume fresh and fully observable telemetry; however, commercial Supervisory Control and Data Acquisition (SCADA) systems routinely encounter transmission latency, packet serialization dropouts, and aggregator-boundary pitch unobservability. Under degraded telemetry, the current operating regime of a turbine ceases to be directly observable and becomes a latent state. Across multi-seed benchmarks on the 134-turbine WTB commercial plant and multi-year evaluations over three European wind facilities, this paper investigates whether machine learning provides universal superiority or conditional utility under information degradation.

Under pristine telemetry, deterministic physical rules achieve the lowest reserve-screening surrogate cost among all evaluated approaches (589,535 kW · h at 10-minute dispatch $h = 1$ and 881,367 kW · h at 1-hour dispatch $h = 6$, with $\sim 6.8\%$ violation), outperforming deep neural networks. Under a 60-minute latency contingency, unadapted physical rules undergo catastrophic reliability breakdown, surging violation rates to $24.0\% \pm 1.7\%$ at $h = 1$ and leaving 127.6 MWh of unhedged shortfall. Crucially, when telemetry channels remain observable, simple state-conditional recalibration absorbs 55.3% of this shortage loss ($127.6 \rightarrow 57.0$ MWh) without neural representation learning. Learned representations acquire demonstrable utility only when blade-pitch registers are withheld: by inferring latent operating states from secondary electromechanical and spatio-temporal consequences (active/reactive responses and wake context), learned representations reduce reserve screening cost by 46.7k–68.6k kW · h across latencies and double transition-window recall (0.416 vs. 0.196). Dynamic Mixture-of-Experts routing confers no statistical advantage over unrouted dense baselines ($p = 0.380$), while decoupled modular architectures match tail reliability (9.7% violation). Furthermore, heuristic selective decision abstention fails to outperform random rejection, being Pareto-dominated by uniform reserve margin expansion. These findings establish that machine learning is valuable not as a universal replacement for aerodynamic rules, but as a conditional latent-state inference mechanism when critical physical states become unobservable.

Index Terms—Wind turbine operating reserve; SCADA telemetry degradation; partial observability; latent operating boundary; transmission latency; aerodynamic power curve; reliability breakdown; reserve-risk screening; state-conditional recalibration; MPPT-to-pitch transition.

I. INTRODUCTION

In bulk power system operations, real-time generation shortfalls incur severe asymmetric shortage penalties relative to surplus energy headroom; consequently, conventional symmetric error metrics like root-mean-square error (RMSE) mask catastrophic tail shortfall events that govern operating reserve adequacy in Level-1 pre-dispatch screening [1], [2], [3], [4], [5]. This operational vulnerability is acutely concentrated near the aerodynamic demarcation boundary between Maximum Power Point Tracking (MPPT, Region 2) and active blade-pitch regulation (Region 3). Near rated inflow velocity, under the idealized rated-power regulation approximation, local aerodynamic power sensitivity transitions abruptly from a steep cubic trajectory ($\partial P / \partial v \propto v^2$) toward zero as pitch actuators rotate the blades to shed aerodynamic lift [6], [7]. When turbine operating states shift across this threshold unannounced, deterministic power curves over-extrapolate generation into Region 3, precipitating severe unhedged reserve deficits.

This operational vulnerability is severely exacerbated by telemetry degradation in commercial Supervisory Control and Data Acquisition (SCADA) systems. Industrial SCADA architectures experience gateway serialization overhead, packet queuing backlogs, and intermittent cellular transmission jitter [8], [9]. In operational commercial archives (such as the Kelmarsh wind plant), 99.6% of discrete turbine status transitions occur strictly between standard 10-minute polling boundaries. Furthermore, in third-party aggregator, Virtual Power Plant (VPP), and Transmission System Operator (TSO) triage, primary blade-pitch angle registers are frequently restricted or unobservable across commercial protocol firewalls. When delayed or missing sensor measurements are fed into deterministic physical power equations, dispatch systems clear reserve margins against obsolete operating points. Under such conditions, the fundamental operational uncertainty is not merely: “*What will the future inflow velocity be?*” It is first and foremost: “*Has the turbine already crossed the aerodynamic operating boundary into a different control regime?*” Under degraded telemetry, the turbine operating regime becomes a partially observed latent state.

Existing literature treats physics-based aerodynamic curves and deep neural networks as competing, mutually exclusive paradigms, pursuing either pure data-driven forecasting or

physics-informed regularization [10], [11], [12], [13], [14], [15], [16]. This framing overlooks their distinct operational boundaries under information freshness constraints. Under pristine SCADA telemetry, deterministic aerodynamic quantile curves calibrated to turbine specifications achieve the lowest reserve-screening surrogate cost among all evaluated methods (589,535 kW·h at immediate dispatch $h = 1$ and 881,367 kW·h at 1-hour dispatch $h = 6$, with compliant $\sim 6.8\%$ violation). However, deterministic rules undergo catastrophic reliability breakdown when subjected to telemetry staleness: under a 60-minute synthetic latency contingency stress test [17], [18], clean-calibrated physical violation rates surge to $24.0\% \pm 1.7\%$ at $h = 1$ and $12.20\% \pm 1.25\%$ at $h = 6$, breaching the nominal 10% Newsvendor target ($q^* = 0.90$) and leaving 127.6 MWh of unhedged shortfall exposure.

To resolve this operational failure, we fundamentally reconstruct the role of machine learning around one central scientific question: *When direct aerodynamic state measurements become stale or unavailable, can consequential spatio-temporal SCADA signals recover a soft operating-boundary posterior that remains dependable for asymmetric reserve-shortfall screening?*

Across 5-seed benchmarks on the 134-turbine WTB commercial plant and multi-year cross-farm evaluations across three European wind plants (ENGIE La Haute Borne, Kelmarsh, Penmanshiel), this paper establishes an empirical operational boundary framework delivering three verifiable insights:

1. **Identification of the Physical Reliability Boundary (RQ1):** We expose the failure mechanism of deterministic aerodynamic rules under transmission staleness, demonstrating that cubic sensitivity cliff transitions amplify delayed-inflow errors and surge violation rates to 24.0%. We bound the operational horizon where deterministic physical rules remain dependable.
2. **Disentangling Recalibration from Latent Representation Recovery (RQ2):** We show that under full channel observability, simple state-conditional recalibration absorbs 55.3% of shortage loss (127.6 $\rightarrow$ 57.0 MWh) at lower computational cost than neural networks. Learned representations acquire demonstrable superiority only when blade-pitch registers are withheld, reconstructing hidden operating regimes from electromechanical transients and spatial wake context, reducing reserve screening cost by 46.7k–68.6k kW·h across latencies and doubling transition recall (0.416 vs. 0.196).
3. **Decision Relevance of the Boundary Posterior and Falsification of MoE Routing (RQ3):** We verify that reserve-screening gains stem from embedding latent boundary awareness into shared spatio-temporal representations and residual quantile estimation, rather than from dynamic Mixture-of-Experts (MoE) routing. STGQ-Routed achieves statistical parity with unrouted dense baselines (cost ratio 1.045, $p = 0.380$), while decoupled modular architectures match tail reliability (9.7% violation). Furthermore, we document an informative negative result: heuristic selective decision abstention fails to outperform random rejection,

being Pareto-dominated by uniform reserve expansion.

II. RELATED WORK

A. *From Average Wind-Power Accuracy to Transition-Window Risk*
Spatio-temporal graph neural networks (STGNNs) have advanced multi-step wind plant forecasting by modeling spatial cross-correlations and dynamic wake advection across turbine arrays [10], [11], [12], [13]. Physics-informed extensions integrate engineering wake models and aerodynamic boundary penalties to constrain continuous forecast trajectories [14], [15], [16]. While these architectures reduce plant-wide L_2 error, continuous trajectory minimization obscures discrete turbine operating state transitions between MPPT (Region 2) and blade-pitch regulation (Region 3) that govern spinning reserve adequacy [19], [20], [21].

Dynamic mixture-of-experts (MoE) routing has been proposed to partition complex feature spaces into localized functional sub-models [22], [23], but unconstrained gating lacks physical interpretability and fails to address the latent nature of turbine operating states under telemetry staleness. In power system operations, probabilistic forecasts monetize uncertainty through reserve procurement [1], [2], [24]. This paper addresses an upstream operational bottleneck: screening aerodynamic transition risks under degraded SCADA telemetry prior to market clearing. The resulting reserve evaluation functions strictly as a Level-1 pre-dispatch screening log for balancing-risk triage, abstracting away downstream transmission network power-flow constraints and multi-stage market cashflows.

III. PROBLEM FORMULATION AND LATENT OPERATING-BOUNDARY FRAMEWORK

A. *Level-1 Pre-Dispatch Risk Screening and Asymmetric Decision Problem (PSREI)*

Operating reserve screening functions as a pre-dispatch risk assessment mechanism to mitigate costly real-time generation imbalances [24], [25].

Two-level hierarchical grid dispatch interface: To define the operational boundaries of this work, we formalize a two-level grid dispatch hierarchy: *Level 1* (Local Pre-Dispatch Risk Filtering & Screening Layer, Paper Scope), operating at the wind plant Energy Management System (EMS) or aggregator terminal to ingest SCADA telemetry, forecast spatio-temporal power, diagnose operational regime boundaries under telemetry degradation, and size pre-dispatch reserve allocations mitigating unhedged balancing exposure; and *Level 2* (Grid-Level Physical Clearing & Network-Constrained Dispatch, Downstream Interface), managed by the TSO to execute network-constrained economic dispatch (NCED), unit commitment, and full AC-OPF subject to bus voltage security, transmission thermal limits, and bulk reserve margins. Our formulation is strictly bounded to Level 1 pre-dispatch risk screening proxies, abstracting away transmission network power-flow constraints and wholesale market cashflow settlements.

Penalized Reserve-Shortfall Energy Index (PSREI): For scheduled point forecast $\hat{y}_{i,t}$ and realized generation $y_{i,t}$, over-forecast shortfall is $s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0)$. Sizing upward reserve margin r_b incurs a reserve procurement energy equivalent $r_b \Delta t$ and an asymmetric shortage penalty

$\rho \max(s_{i,t} - r_b, 0) \Delta t$. For penalty ratio ρ , the total operational proxy metric is formalized as the **Penalized Reserve-Shortfall Energy Index (PSREI)** in $\text{kW} \cdot \text{h}$ (evaluated at representative proxy ratio $\rho = 10$, alongside sensitivity evaluations across $\rho \in \{5, 10, 20\}$):

$$C(r_b; \rho) = \sum_{(i,t)} [r_b(i, t) + \rho \max(\hat{y}_{i,t} - y_{i,t} - r_b(i, t), 0)] \Delta t. \quad (1)$$

PSREI is a decision-oriented risk surrogate targeting pre-dispatch shortage mitigation rather than financial market settlement. Differentiating $\mathbb{E}[C(r_b; \rho)]$ with respect to reserve margin r_b yields the first-order optimality condition $\partial \mathbb{E}[C]/\partial r_b = 1 - \rho \Pr[s_{i,t} > r_b] = 0$, giving the classical Newsvendor critical fractile $q^*(\rho) = 1 - 1/\rho$ [1]. For our representative proxy ratio $\rho = 10$, $q^*(10) = 0.90$, establishing a nominal 10% violation target.

Loss geometry and pinball quantile head: Conventional point forecasting optimizes symmetric quadratic loss $\mathcal{L}_{L_2}(y, \hat{y}) = (y - \hat{y})^2$, whose Bayes-optimal predictor converges to conditional expectation $\mathbb{E}[y | \mathbf{x}]$. In bounded wind power regimes with non-Gaussian transition densities, the conditional mean systematically departs from upper conditional quantiles, while balancing operations ($\rho = 10$) penalize generation shortfalls ten times more heavily than surplus headroom. Upward reserve margin $\hat{r}_{i,t}$ required to cover shortfall residual $s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0)$ is estimated via asymmetric pinball loss:

$$\mathcal{L}_q(s, \hat{r}) = (s - \hat{r}) (q - \mathbf{1}[s < \hat{r}]) = \begin{cases} q(s - \hat{r}), & s \geq \hat{r}, \\ (1 - q)(\hat{r} - s), & s < \hat{r}. \end{cases} \quad (2)$$

Setting $q = q^* = 0.90$ aligns training with the Newsvendor condition: Bayes-optimal prediction converges to conditional shortfall quantile $\hat{r}^* = Q_{0.90}(s | \mathbf{x})$, targeting a nominal 10% exceedance probability [1], [2].

PCC portfolio smoothing & multi-horizon dispatch: Power delivered to the bulk grid is metered at the Point of Common Coupling (PCC) bus: $P_{\text{farm},t} = \sum_{i=1}^M P_{i,t}$ and $R_{\text{farm},t} = \sum_{i=1}^M R_{i,t}$. Aggregating decentralized turbine power flows at the PCC bus unlocks spatial portfolio diversification: localized high-frequency aerodynamic turbulence cancels out across the array, leaving boundary regime shifts as the primary source of unhedged reserve risk. We evaluate two distinct operational horizons: (1) **Immediate dispatch** ($h = 1$, 10-min ahead), representing real-time economic dispatch (RTED) dominated by autocorrelation; and (2) **Operational dispatch** ($h = 6$, 1-hour ahead), representing intra-day clearing and storage scheduling where spatial wake dynamics and weather transitions are paramount.

B. Latent Operating Regimes: MPPT, Pitch Regulation, and Demarcation Boundaries

Let $G = (V, E)$ denote a turbine array spatial graph with N nodes. For each turbine $i \in V$ at discrete time step t , the system observes a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predicts active power $y_{i,t} \in \mathbb{R}$. Given observation history H and forward horizon P , the forecast mapping is:

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathcal{F}(\mathbf{X}_{t-H+1:t}, \mathcal{A}_{t-H+1:t}), \quad (3)$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H \times N \times F}$ and $\mathcal{A}_{t-H+1:t}$ denotes a time-varying directed graph sequence (WTB) or static graph repeated over time (ERA5). We enforce a strict causal boundary: all input features, graph adjacency structures, and supervisory anchors depend solely on telemetry available at or before dispatch anchor t , with target trajectories commencing strictly at $t+1$. In WTB, instantaneous active power $\text{Patv}_{i,t}$ serves solely as an anchor-time operational status indicator, preventing look-ahead leakage.

At any dispatch time t , turbine i operates in a discrete aerodynamic control regime governed by local inflow wind speed $w_{i,t} = \text{Wspd}_{i,t}$ and collective blade pitch angle $\bar{p}_{i,t} = \frac{1}{3} \sum_{k=1}^3 p_{i,t}^{(k)}$:

$$Z_{i,t} = \begin{cases} 0, & w_{i,t} < u_{\text{idle}} \quad (\text{idle}), \\ 1, & u_{\text{idle}} \leq w_{i,t} \leq u_{\text{rated}}, \bar{p}_{i,t} < p_{\text{th}} \quad (\text{MPPT}), \\ 2, & w_{i,t} > u_{\text{rated}}, \bar{p}_{i,t} \geq p_{\text{th}} \quad (\text{pitch-regulated}), \\ 3, & \text{otherwise} \quad (\text{transitional/ambiguous}). \end{cases} \quad (4)$$

Thresholds are calibrated to turbine specifications: WTB (134 turbines: $u_{\text{idle}} = 3.0$ m/s, $u_{\text{rated}} = 10.5$ m/s, $p_{\text{th}} = 2.0^\circ$), Kelmarsh (6 MM92 turbines: 3.0 m/s, 12.5 m/s, 1.0°), Penman-shiel (15 MM82 turbines: 3.0 m/s, 14.5 m/s, 1.0°), and ENGIE La Haute Borne (4 MM82 turbines: 3.0 m/s, 14.5 m/s, 1.0°). Only well-defined regimes ($Z_{i,t} \in \{0, 1, 2\}$) provide direct supervisory alignment; transitional samples are masked via $M_{i,t} = \mathbf{1}[Z_{i,t} \neq 3]$.

Under pristine and complete telemetry, $\bar{p}_{i,t}$ and $w_{i,t}$ are known, so $Z_{i,t}$ is directly observable. However, when transmission latency $\tau > 0$ delays the SCADA stream or when blade-pitch registers are restricted across aggregator interfaces, $Z_{i,t}$ ceases to be directly observable and becomes a **partially observed latent state**. The central inferential object of the operational system is the soft operating-boundary posterior:

$$p(Z_{i,t} | \mathbf{X}_{\leq t}), \quad (5)$$

which must be inferred from historical and spatial consequence channels.

C. Aerodynamic Power Conversion and the Stale-State Control Cliff

In variable-speed wind turbine aerodynamics, mechanical power captured from inflow air is governed by $P_{\text{mech}}(t) = \frac{1}{2} \rho_{\text{air}} \pi R^2 C_p(\lambda(t), \beta(t)) v^3(t)$, where $\rho_{\text{air}} \approx 1.225$ kg/m³, R is rotor radius, $v(t)$ is wind speed, $\beta(t)$ is blade pitch angle, and $\lambda(t) = \omega_r(t)R/v(t)$ is tip-speed ratio. Standard aeroelastic turbine models approximate $C_p(\lambda, \beta)$ via non-linear empirical polynomials [6], [7] (formalized in Supplementary Appendix-A).

Turbine dynamics are partitioned by rated wind speed u_{rated} : in Region 2 (MPPT, $u_{\text{idle}} \leq v \leq u_{\text{rated}}$), blades maintain optimal fine pitch $\beta \approx 0^\circ$ yielding cubic power growth $P(v) \propto v^3$; in Region 3 (Pitch regulation, $u_{\text{rated}} < v \leq u_{\text{cut-out}}$), the pitch controller rotates blades to shed lift, clamping electrical output at nameplate capacity P_{rated} . In idealized aerodynamic representations, this transition exhibits a localized slope discontinuity [6]:

$$\left. \frac{\partial P}{\partial v} \right|_{v \rightarrow u_{\text{rated}}^-} = \frac{3}{2} \rho_{\text{air}} \pi R^2 C_{p,\text{max}} u_{\text{rated}}^2 \gg 0, \quad \left. \frac{\partial P}{\partial v} \right|_{v \rightarrow u_{\text{rated}}^+} \approx 0. \quad (6)$$

When SCADA telemetry experiences latency $\tau > 0$, deterministic aerodynamic curves evaluate stale state tuples $(v_{t-\tau}, \beta_{t-\tau})$. If the turbine crosses from Region 2 into Region 3 during the latency interval ($Z_{t-\tau} = 1$ while $Z_t = 2$), the delayed pitch register remains at fine pitch $\beta_{t-\tau} \approx 0^\circ$. Consequently, the deterministic aerodynamic rule evaluates an obsolete operating point, extrapolating power along the steep cubic slope:

$$\hat{P}_{\text{phys}}(t) = \frac{1}{2} \rho_{\text{air}} \pi R^2 C_p(\lambda(t), 0^\circ) v^3(t) \gg P_{\text{rated}}. \quad (7)$$

Because the actual turbine output is clamped at P_{rated} , this stale evaluation produces a severe over-forecast error $\hat{P}_{\text{phys}}(t) - P_{\text{rated}} \gg 0$. In pre-dispatch reserve screening with asymmetric penalty ratio $\rho = 10$, this unhedged over-forecast produces catastrophic shortage penalties, driving empirical violation rates to $24.0\% \pm 1.7\%$ under Delay-6. This establishes the deterministic physical power-curve baseline and exposes its structural failure mode under stale telemetry.

D. The Non-Neural Baseline: State-Conditional Quantile Recalibration

Before introducing neural representation learning, it is necessary to establish the non-neural limit of adaptation. To evaluate whether distribution drift induced by latency can be absorbed by lightweight non-neural statistical correction, we formalize state-conditional quantile recalibration.

Given delayed telemetry $\mathbf{s}_{i,t-\tau} = (w_{i,t-\tau}, \bar{p}_{i,t-\tau})$, we partition the state space into K operational bins $\{\mathcal{B}_k\}_{k=1}^K$ defined by cut-in, rated wind speed, and pitch thresholds ($Z_{i,t-\tau} = k$). To eliminate data leakage, empirical quantile reserve margins $\hat{r}(k)$ are calibrated strictly on held-out validation residuals $\mathcal{D}_k^{\text{val}}$:

$$\hat{r}(k) = \inf \left\{ r \in \mathbb{R} : \frac{1}{|\mathcal{D}_k^{\text{val}}|} \sum_{(i,t) \in \mathcal{D}_k^{\text{val}}} \mathbf{1}[\hat{y}_{i,t}^{\text{phys}} - y_{i,t} \leq r] \geq q^* \right\}, \quad (8)$$

where $\mathcal{D}_k^{\text{val}} = \{(i, t) \in \mathcal{D}_{\text{val}} : \mathbf{s}_{i,t-\tau} \in \mathcal{B}_k\}$ and $q^* = 1 - 1/\rho = 0.90$.

We formalize two distinct calibration protocols: (1) *Clean-Frozen Calibration*, where reserve margins $\hat{r}^{\text{clean}}(k)$ are calibrated exclusively on nominal clean validation data ($\tau = 0$) and strictly frozen prior to deployment, quantifying unadapted vulnerability under latency ($\tau > 0$); and (2) *Condition-Matched Validation Calibration*, where for a known delay τ , reserve margins $\hat{r}(k; \tau)$ are re-estimated on held-out validation residuals undergoing the identical delay τ , establishing a non-neural benchmark that absorbs distribution drift under full channel observability. Observing delayed telemetry in bin $\mathcal{B}_k$ yields reserve allocation $\hat{r}_{i,t} = \hat{r}(k)$.

Crucially, state-conditional recalibration is strictly an **information-preserving correction**: it recalibrates the residual mapping conditional on observed state bins, but it cannot synthesize missing physical state information when critical channels are withheld. When primary blade-pitch telemetry $\bar{p}_{i,t}$ is withheld, the physical state cannot be binned accurately, causing recalibration to deteriorate.

E. Latent State Recovery from Spatio-Temporal Consequence Channels

Physical dynamic wake graph: When primary SCADA telemetry degrades or pitch registers are withheld, spatial coupling across turbine arrays provides physical redundancy to infer local operating regimes [11], [26]. For WTB, we construct a time-varying directed wake graph $\mathcal{A}_t$: candidate turbine pairs (i, j) are filtered by physical distance ($d_{ij} \leq d_{\text{max}}$), activated when upstream turbine j lies within an aerodynamic wake cone aligned with instantaneous local wind direction θ_t (half-angle $\alpha = 25^\circ$), weighted by streamwise decay $\exp(-d_{\parallel}/\sigma_x)$ and cross-stream decay $\exp(-d_{\perp}^2/2\sigma_y^2)$, and pruned to the K strongest inbound wake neighbors [14], [16]. This defines an instantaneous aerodynamic wake score $s_{i,t}^{\text{wake}}$, providing spatial context even during anemometer corruption.

Directed-diffusion GRU encoder: With 110k parameters and an average single-window execution time of 3.8 ms in desktop benchmarking, the encoder features a compact parameter footprint suitable for local substation computing envelopes. Two directed diffusion blocks aggregate self, inbound, and outbound messages across each graph snapshot:

$$\mathbf{X}_t^{(\ell+1)} = \text{LN}(\mathbf{W}_{\text{self}}^{(\ell)} \mathbf{X}_t^{(\ell)} + \mathbf{W}_{\text{in}}^{(\ell)} \text{Agg}_{\text{in}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{W}_{\text{out}}^{(\ell)} \text{Agg}_{\text{out}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{R}^{(\ell)} \mathbf{X}_t^{(\ell)}), \quad (9)$$

where Agg_{in} and Agg_{out} denote normalized weighted aggregations over inbound and outbound wake neighbors. A GRU then extracts node-level representations $\mathbf{h}_{i,t}$.

Cross-sensor electromechanical consequence mechanism: When blade-pitch telemetry $\bar{p}_{i,t}$ suffers packet loss, transmission latency, or protocol unobservability across aggregator boundaries, the dynamic graph encoder leverages secondary consequence channels to infer the latent operating regime $p(Z_{i,t} | \mathbf{X}_{\leq t})$: 1. *Active power transients* ($\text{Patv}_{i,t}$): Divergence between active power output and inflow velocity reflects aerodynamic torque shedding; 2. *Electrical and reactive responses*: Variations in terminal voltage and reactive power dynamics signal transition-region control action; 3. *Spatial wake context*: Dynamic wake advection across upstream and downstream turbines provides spatial corroboration of boundary crossing.

In nominal operation, the boundary head ingests $\mathbf{a}_{i,t}^{\text{WTB}} = [\text{wspd}_{i,t}, \text{Pab_mean}_{i,t}, s_{i,t}^{\text{wake}}, \text{Patv}_{i,t}]$. Under pitch-withheld conditions, $\text{Pab_mean}_{i,t}$ is strictly excluded, and the representation learns to reconstruct $Z_{i,t}$ from secondary consequences alone.

F. Reserve-Tail Estimation and Residual Quantile Sizing

To shield grid operations from unhedged balancing exposure, reserve margins must target the asymmetric shortfall distribution rather than symmetric point forecasts. The framework maintains an explicit mathematical separation among three distinct inferential objects: 1. **Posterior operating-state inference**: $\pi_{i,t}^{\text{pitch}} = p(Z_{i,t} = 2 | \mathbf{X}_{\leq t})$, estimating the probability of active blade-pitch regulation; 2. **Point power trajectory forecasting**: $\hat{\mathbf{y}}_{i,t+1:t+P} = \mathbb{E}[\mathbf{y}_{i,t+1:t+P} | \mathbf{X}_{\leq t}]$, providing the scheduled dispatch profile; 3. **Reserve-tail quantile estimation**: $\hat{r}_{i,t} = Q_{q^*}(s_{i,t} | \mathbf{X}_{\leq t})$, sizing upward reserve to cover asymmetric shortfall exposure $s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0)$ at

News vendor critical fractile $q^* = 1 - 1/\rho = 0.90$.

Directly estimating extreme quantiles on raw power trajectories via uncalibrated neural pinball regression introduces severe tail conservatism: without hard physical saturation bounds, direct neural quantile heads over-procure reserve margins by $> 1.6\text{M kW}$ (violation rate $< 1.2\%$, Table~I), incurring prohibitive capacity holding costs. To resolve this pathology, reserve-tail estimation is decoupled: point trajectories $\hat{\mathbf{y}}$ anchor nominal expectations, while upward reserve margins $\hat{r}_{i,t}$ are estimated conditionally on forecast residuals:

$$\hat{r}_{i,t} = \text{MLP}_{\text{res}}([\mathbf{h}_{i,t}; \mathbf{x}_{\text{anchor}}]), \quad \text{minimized under } \mathcal{L}_{q^*}(s_{i,t}, \hat{r}_{i,t}), \quad (10)$$

or binned empirically over held-out validation residuals conditioned on the latent state posterior $\pi_{i,t}^{\text{pitch}}$. This guarantees that reserve sizing remains calibrated to actual residual dispersion across distinct operating regimes.

G. Architecture Variants as Mechanism Ablations

To isolate the causal mechanism driving reserve-screening performance and determine whether dynamic Mixture-of-Experts (MoE) routing provides genuine operational value over unrouted dense or modular representations, we formulate three architectural variants under matched parameter capacity: 1. **STGQ-Modular** (Decoupled Canonical Architecture): A modular architecture pairing frozen spatio-temporal representations $\mathbf{h}_{i,t}$ with a dedicated residual quantile head $\hat{r}_{i,t} = \text{MLP}_{\text{res}}([\mathbf{h}_{i,t}; \mathbf{x}_{\text{anchor}}])$ or posterior-binned empirical quantiles. Decoupling representation learning from quantile regression prevents auxiliary decision loss from distorting spatio-temporal feature extraction; 2. **STGQ-Dense** (Unrouted Joint Baseline): An unrouted baseline coupling spatial graph embeddings $\mathbf{h}_{i,t}$ directly to a shared linear projection for point forecasts $\hat{\mathbf{y}}_{i,t+1:t+P}$ and a joint dense quantile head, omitting dynamic routing; 3. **STGQ-Routed** (Dynamic MoE Gating Comparator): An architectural comparator evaluating dynamic Mixture-of-Experts (MoE) routing [22], [23], where gating logits $\mathbf{z}_{i,t} = \text{MLP}_{\text{gate}}([\mathbf{h}_{i,t}; \mathbf{a}_{i,t}])$ compute soft routing weights $\mathbf{g}_{i,t} = \text{softmax}(\mathbf{z}_{i,t}/\tau_{\text{gate}})$ to combine specialized expert outputs: $\hat{\mathbf{y}}_{i,t+1:t+P} = \sum_{e=1}^E g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t})$.

The joint models are trained with a multi-task objective combining multi-step trajectory prediction loss $\mathcal{L}_{\text{pred}}$ with auxiliary physical regularizers:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{force}} \mathcal{L}_{\text{force}}, \quad (11)$$

where $\mathcal{L}_{\text{align}}$ anchors routing/classification logits to declared regimes $Z_{i,t}$ on unmasked samples ($M_{i,t} = 1$) using train-only inverse-frequency weights; $\mathcal{L}_{\text{force}}$ sharpens discrimination between MPPT and pitch regulation; $\mathcal{L}_{\text{bal}}$ is an expert-balancing regularizer used exclusively for the routed comparator to prevent expert collapse; and $\mathcal{L}_{\text{aux}}$ with $\mathcal{L}_{\text{smooth}}$ enforce spatial wake consistency.

Explicit Methodological Notice: We emphasize that $\mathcal{L}_{\text{align}}$ acts strictly as an **operating-regime prior** embedded into the shared representation space; $\mathcal{L}_{\text{bal}}$ serves solely as a routing regularizer. **The neural backpropagation pipeline contains no quantile loss, pinball penalty, or News vendor decision loss.** Model checkpoint selection is strictly governed by valida-

tion RMSE, preventing auxiliary penalties from distorting prediction accuracy. All quantile reserve margins are calibrated on held-out validation residuals or estimated by the decoupled residual quantile head. Therefore, the learned posterior is a *jointly represented boundary posterior* rather than an end-to-end posterior calibrated directly under dispatch decision loss.

IV. CASE STUDY CONFIGURATION AND OPERATIONAL CONSTRAINTS

A. Datasets and Preprocessing

Empirical evaluation spans two contrasting observability settings: the 134-turbine WTB commercial wind plant benchmark and the ERA5 thermodynamic contrast patch. WTB (KDD Cup 2022) comprises 134 wind turbines and 245 days of continuous 10-minute SCADA telemetry ($T = 35,280$, $N = 134$, $F = 11$) [27], where turbulent inflow, wake propagation, and turbine pitch regulation interact nonlinearly, rendering the MPPT-to-pitch transition indirectly observable. Missing inputs are forward-filled per turbine and mean-imputed; non-positive power values are masked out of supervision. ERA5 covers three archived months on a 16×16 spatial grid ($T = 2208$, $N = 256$, eight features) [28], where surface sensible heat flux directly marks convective regimes. Both datasets adopt an observation history of $H = 36$ and a prediction horizon of $P = 24$ under chronological train/validation/test splits (180/30/35 days for WTB; 1325/441/442 frames for ERA5). In addition, multi-year SCADA records from three European commercial wind plants (ENGIE La Haute Borne, Kelmarsh, Penmanshiel) provide diverse turbine architectures for cross-farm evaluation.

B. Controlled Telemetry Degradation Protocols and Realism Mapping

To stress-test reserve screening resilience under adverse communication conditions, we evaluate controlled counterfactual telemetry degradation protocols alongside sensor perturbation controls across all five seeds (201–205): (1) **Tier 1 (Nominal SCADA Polling, 0–10 min, Clean)**: fresh telemetry on the 10-min SCADA grid (IEC 61400-25); (2) **Tier 2 (Plausible Degraded Telemetry, 10–30 min, Delay-1 to Delay-3)**: communication latencies induced by substation ring-buffer backlogs and lossy cellular retransmissions [9]; and (3) **Tier 3 (Contingency Stress-Test Upper Envelope, 60 min, Delay-6)**: a 6-step (60 min) synthetic transmission lag evaluated strictly as an upper contingency stress envelope (simulating gateway failures or cyber-physical storms) [17], [18] to establish the asymptotic breakdown boundary of unadapted physical rules. Sensor perturbation controls comprise additive Gaussian noise ($\sigma_v = 1.0 \text{ m/s}$, $\sigma_\beta = 2.0^\circ$) and a two-state Markov-Gilbert burst dropout chain ($p_{GB} = 0.08$, $p_{BB} = 0.75$, $d \leq 6$) simulating transient wireless dropouts [9], [18].

Fig.~1 illustrates the causal decision failure mechanism and the three-tier operational recovery hierarchy across telemetry degradation regimes.

C. Pre-Dispatch Risk Screening Framework and Resampling Hierarchy

All neural architectures share an identical training protocol using AdamW with early stopping on validation RMSE, gradient clipping, and mixed-precision execution on a single GPU across five strict-mask random seeds (201–205). Strong

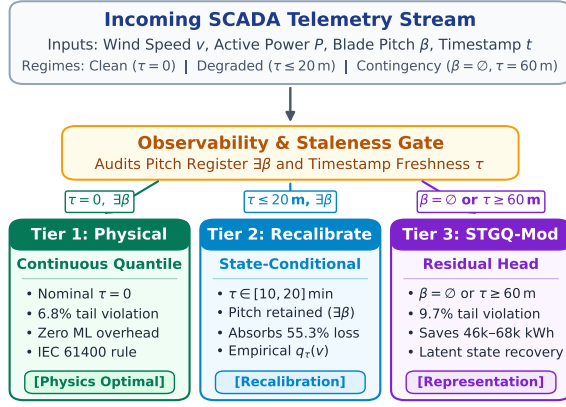


Fig. 1. Causal decision failure and information recovery hierarchy under SCADA telemetry degradation. Panel A: Physical mechanism and staleness breakdown at rated cut-off, showing how telemetry staleness turns the operating regime Z_t into a latent variable, precipitating cubic extrapolation errors and severe reserve shortfalls (24.0% violation, 127.6 MWh deficit). Panel B: Three-tier information-centric recovery hierarchy, routing fresh telemetry to Tier 1 deterministic physical rules, observable delay to Tier 2 state-conditional recalibration, and pitch-withheld/contingency telemetry to Tier 3 learned latent representations (STGQ-Modular). Panel C: Telemetry mechanism falsification and negative results, showing that dynamic MoE routing yields no advantage over unrouted dense/modular architectures ($p = 0.380$) and heuristic selective abstention fails to beat random rejection.

baselines (Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, TiDE) repeat across seeds 201–205 (30 completed runs under identical evaluation caches). To eliminate data leakage, the experimental pipeline strictly enforces a sequential three-split dataflow: $\mathcal{D}_{\text{train}} \xrightarrow{\text{train } \theta} \mathcal{D}_{\text{val}} \xrightarrow{\text{calibrate } \hat{r}(k)} \text{Freeze } \{\theta, \hat{r}\} \xrightarrow{\text{causal replay}} \mathcal{D}_{\text{test}}$. Quantile offsets $\hat{r}(k)$ are estimated strictly on held-out validation residuals $\mathcal{D}_{\text{val}}$ and frozen prior to test replay.

We establish a dual-level resampling hierarchy: (1) *Model Initialization Robustness*, reporting mean $\pm$ standard deviation across five random seeds (201–205); and (2) *Operational Time-Series Uncertainty*, evaluating operational risk metrics via a day-block bootstrap over the 35 test days (1,000 resamples of 144-step daily blocks, preserving diurnal cycles and serial autocorrelation) to construct empirical 95% confidence intervals and paired difference distributions. Evaluation spans: point accuracy (MAE/RMSE, switch-window RMSE within ± 1.0 m/s of rated wind speed); physical regime alignment (N-MI, ARI, gate entropy); and operational reserve consequence (PSREI reserve-shortfall surrogate cost, empirical violation rate, unhedged shortage MWh).

V. EMPIRICAL FINDINGS AND HYPOTHESIS TESTING

A. Result 1: Physics under Fresh Telemetry (RQ1)

Table~I summarizes the multi-day replay dispatch benchmark for 1-hour ahead operational dispatch ($h = 6$) across 134 turbines over the 35-day test split under nominal clean telemetry. Table~II reports seed-paired hypothesis tests and mechanism ablations across operational regimes, along with 95% day-block cluster bootstrap confidence intervals.

The empirical evidence directly challenges the presumption that deep neural architectures universally outperform physical models. Under pristine telemetry ($\tau = 0$), Continuous Physical Quantile achieves the lowest surrogate reserve screening cost

among all evaluated methods ($589,535 \pm 77,390$ kW · h at immediate dispatch $h = 1$ and $881,367 \pm 95,741$ kW · h at 1-hour dispatch $h = 6$, with compliant $\sim 6.8\%$ violation). It substantially outperforms both unrouted dense networks (STGQ-Dense: 991,181 kW · h) and dynamically routed architectures (STGQ-Routed: 959,027 kW · h). As confirmed in Table~II, the seed-paired difference between Continuous Physical Quantile and STGQ-Routed is $-77,659$ kW · h, with a 95% block bootstrap confidence interval strictly excluding zero ($[-106,175, -49,143]$).

This finding establishes that when inflow wind speed and blade pitch angle are fresh and directly observable, deterministic aerodynamic equations provide optimal reserve screening. Neural networks, lacking hard physical saturation bounds, introduce unnecessary variance under nominal telemetry.

B. Result 2: State-State Boundary Failure (RQ1)

While deterministic physical rules excel under fresh data, Table~I exposes their **catastrophic reliability breakdown** under telemetry staleness. Under a 60-minute latency contingency stress test (Delay-6), deterministic physical curves evaluate stale state tuples $(v_{t-\tau}, \beta_{t-\tau})$. Near rated wind speed, the localized slope discontinuity (6) amplifies delayed-inflow errors: turbines that have crossed into active blade-pitch regulation ($Z_t = 2$) are evaluated with obsolete fine pitch ($\beta_{t-\tau} \approx 0^\circ$), over-extrapolating power output along the steep cubic trajectory (7).

Consequently, unadapted physical quantile violation rates surge from 6.81% to $24.02\% \pm 1.69\%$ at $h = 1$ and $12.20\% \pm 1.25\%$ at $h = 6$, breaching the nominal 10% Newsvendor target ($q^* = 0.90$). This breakdown leaves 127.55 MWh of unhedged generation shortfall across the array. This delineates the exact operational reliability boundary of deterministic aerodynamic rules: when transmission latency delays state observation, physical rules cease to provide dependable reserve screening.

C. Result 3: Full Observability Drift Is Resolved by Recalibration (RQ2)

When telemetry channels remain observable under transmission delays, condition-matched state-conditional recalibration (8) adjusts quantile margins using held-out validation residuals undergoing the identical delay. As shown in Table~III, state-conditional recalibration absorbs 55.3% of the shortage loss ($127.55 \rightarrow 57.01$ MWh), restoring fleet violation to 9.46% and achieving a surrogate reserve cost of 1,228,609 kW · h at $h = 1$. In fact, recalibrated physical curves outperform all neural models under full observability (STGQ-Routed: 1,274,047 kW · h; STGQ-Modular: 1,283,855 kW · h).

To establish the operational limits of recalibration under communication drift, we evaluate a mismatched calibration matrix across test latencies $\tau_{\text{test}} \in [0, 60]$ min and calibration horizons $\tau_{\text{cal}} \in [0, 60]$ min (Supplementary Table~A111). Recalibrating at $\tau_{\text{cal}} = 10$ min withstands latency drift up to $\tau_{\text{test}} = 20$ min while maintaining compliant tail coverage (8.2%–9.1% violation). However, recalibration breaks down when staleness surges to $\tau_{\text{test}} = 60$ min (13.8% violation). This demonstrates that under full channel observability, distribution drift alone does not justify deep representation learning:

TABLE I

CROSS-SEED MULTI-REGIME PRE-DISPATCH RESERVE SCREENING BENCHMARK UNDER CLEAN-CALIBRATED PROTOCOL ($h = 6$, 1-HOUR AHEAD DISPATCH, 5 SEEDS 201–205). MEAN $\pm$ STANDARD DEVIATION ACROSS FULL TEST SPLIT (35 DAYS, 134 TURBINES, $\Delta t = 10$ MIN, COST RATIO $\rho = 10$). ALL QUANTILE BINS AND MULTIPLIERS CALIBRATED EXCLUSIVELY ON NOMINAL CLEAN VALIDATION DATA.

Regime	Evaluated Model	Penalized Reserve-Shortfall Energy (kW·h, Proxy at $\rho = 10$)	Violation Rate	Reserve (kW)	Shortage (kW·h)	Pinball Loss
Clean	Continuous Physical Quantile	881,367 $\pm$ 95,741	6.81% $\pm$ 1.13%	653,738 $\pm$ 72,557	22,763 $\pm$ 4,013	31.41 $\pm$ 2.66
	Frozen-Embedding Direct Quantile MLP [‡]	1,649,721 $\pm$ 285,682	0.58% $\pm$ 0.26%	1,630,572 $\pm$ 293,346	1,915 $\pm$ 958	64.62 $\pm$ 12.05
	Global Quantile	998,988 $\pm$ 107,360	6.46% $\pm$ 1.31%	771,304 $\pm$ 95,661	22,768 $\pm$ 4,854	36.50 $\pm$ 3.41
	STGQ-Dense	991,181 $\pm$ 110,211	6.54% $\pm$ 1.04%	761,480 $\pm$ 94,244	22,970 $\pm$ 4,121	36.16 $\pm$ 3.58
	STGQ-Routed	959,027 $\pm$ 103,290	7.10% $\pm$ 1.17%	707,249 $\pm$ 91,305	25,178 $\pm$ 4,816	34.77 $\pm$ 3.16
	Missingness-Aware GBDT	1,238,603 $\pm$ 112,051	4.50% $\pm$ 0.73%	1,085,168 $\pm$ 118,015	15,343 $\pm$ 2870	46.85 $\pm$ 3.48
Delay-6	Continuous Physical Quantile	1,136,466 $\pm$ 84,101	12.20% $\pm$ 1.25% [†]	662,335 $\pm$ 73,483	47,413 $\pm$ 3,091	38.85 $\pm$ 2.06
	Frozen-Embedding Direct Quantile MLP [‡]	1,682,251 $\pm$ 284,323	1.20% $\pm$ 0.52%	1,651,979 $\pm$ 297,715	3,027 $\pm$ 1,594	62.15 $\pm$ 11.70
	Global Quantile	1,180,603 $\pm$ 91,754	10.85% $\pm$ 1.55% [†]	780,916 $\pm$ 96,853	39,969 $\pm$ 6,087	40.73 $\pm$ 2.43
	STGQ-Dense	1,179,859 $\pm$ 95,268	11.04% $\pm$ 1.22% [†]	773,103 $\pm$ 94,462	40,676 $\pm$ 4,788	40.70 $\pm$ 2.66
	STGQ-Routed	1,163,593 $\pm$ 91,299	11.82% $\pm$ 1.64% [†]	720,015 $\pm$ 92,196	44,358 $\pm$ 7,341	40.01 $\pm$ 2.54
	Missingness-Aware GBDT	1,320,003 $\pm$ 93,135	8.36% $\pm$ 0.97%	1,030,858 $\pm$ 114,266	28,915 $\pm$ 4,469	46.68 $\pm$ 2.45
Noise	Continuous Physical Quantile	998,273 $\pm$ 106,253	7.59% $\pm$ 0.95%	746,697 $\pm$ 82,474	25,158 $\pm$ 3,968	33.97 $\pm$ 2.78
	Frozen-Embedding Direct Quantile MLP [‡]	1,639,909 $\pm$ 242,922	0.80% $\pm$ 0.28%	1,618,439 $\pm$ 249,374	2,147 $\pm$ 953	60.62 $\pm$ 9.85
	Global Quantile	1,053,552 $\pm$ 114,150	7.48% $\pm$ 1.48%	802,361 $\pm$ 99,513	25,119 $\pm$ 5,822	36.26 $\pm$ 3.24
	STGQ-Dense	1,048,083 $\pm$ 115,181	7.63% $\pm$ 1.13%	792,472 $\pm$ 92,391	25,561 $\pm$ 4,643	36.04 $\pm$ 3.28
	STGQ-Routed	1,027,077 $\pm$ 114,317	7.62% $\pm$ 1.09%	769,864 $\pm$ 106,672	25,721 $\pm$ 3,963	35.16 $\pm$ 3.19
	Missingness-Aware GBDT	1,314,640 $\pm$ 129,158	4.66% $\pm$ 0.80%	1,160,329 $\pm$ 137,650	15,431 $\pm$ 3,099	47.11 $\pm$ 3.87
Markov	Continuous Physical Quantile	907,871 $\pm$ 96,820	7.20% $\pm$ 1.15%	663,875 $\pm$ 73,821	24,400 $\pm$ 4,106	31.59 $\pm$ 2.61
	Frozen-Embedding Direct Quantile MLP [‡]	1,680,339 $\pm$ 289,602	0.57% $\pm$ 0.25%	1,660,135 $\pm$ 298,104	2,020 $\pm$ 992	64.38 $\pm$ 12.00
	Global Quantile	1,030,131 $\pm$ 108,592	6.87% $\pm$ 1.39%	785,305 $\pm$ 97,397	24,483 $\pm$ 5,104	36.78 $\pm$ 3.35
	STGQ-Dense	1,020,078 $\pm$ 112,195	6.90% $\pm$ 1.07%	775,899 $\pm$ 96,235	24,418 $\pm$ 4,246	36.35 $\pm$ 3.57
	STGQ-Routed	990,625 $\pm$ 104,487	7.51% $\pm$ 1.22%	720,858 $\pm$ 92,754	26,977 $\pm$ 5,183	35.10 $\pm$ 3.10
	Missingness-Aware GBDT	1,261,941 $\pm$ 112,754	4.88% $\pm$ 0.75%	1,094,730 $\pm$ 120,612	16,721 $\pm$ 3,298	46.62 $\pm$ 3.40

[†]Exceeds nominal 10% violation target ($q^* = 0.90$). Bold numbers denote lowest surrogate screening metric. **Protocol Attribution:** Reports the *Clean-Calibrated Operational Protocol* at $h = 6$ (thresholds frozen on clean validation data); Continuous Physical Quantile collapses to 12.20% violation under Delay-6. For the *State-Conditional Validation-Calibrated Protocol* at immediate horizon ($h = 1$), see Section V-B, where clean-calibrated physical rules collapse to 24.0% violation under Delay-6 while state-conditional calibration mitigates violations to 10.7% (physical) and 10.6% (joint model). Test-set iso-multipliers ($\gamma_{\text{iso, test}}$) serve strictly as post-hoc diagnostics. [‡]Frozen-Embedding Direct Quantile MLP evaluates naive uncalibrated pinball regression on frozen representations, exhibiting severe tail conservatism (>1.63 M kW reserve, $<1.2\%$ violation). In contrast, the calibrated residual architecture (STGQ-Modular, Section V-F, Table III) and calibrated modular baselines (Cascaded Frozen MLP 15.528M $\pm$ 1.599M vs STGQ-Routed 16.065M with no statistically detectable difference) resolve this pathology.

simple state-conditional recalibration captures the vast majority of recoverable value.

D. Result 4: Pitch Unobservability Demands Latent State Representation (RQ2)

The indispensable role of machine learning emerges when blade-pitch telemetry is withheld across aggregator boundaries (no_pitch, Table~III). Under pitch unobservability, physical rules and recalibration schemes lose direct state awareness, inflating recalibrated screening penalties from 824,393 kW · h at $\tau = 0$ to 1,378,900 kW · h at $\tau = 60$ min.

TABLE III

MULTIDIMENSIONAL OPERATIONAL PROGRESSION ACROSS TELEMETRY LATENCY $\tau \in [0, 60]$ MIN AND PITCH OBSERVABILITY ($h = 1$, 5-SEED MEAN ON WTB; NOMINAL $\tau = 0$ BASELINE 593,258 kW · h ALIGNS WITHIN 0.6% OF THE PRIMARY FROZEN 589,535 kW · h BENCHMARK). PENALIZED RESERVE ENERGY IN kW·H ($\rho = 10$).

τ (min)	Pitch	Clean Phys. Metric (Viol.)	Recal. Phys. Metric (Viol.)	STGQ-Mod. Metric (Viol.)	STGQ-Rout. Metric (Viol.)	Operational Regime
0	All	593,258 (7.3%)	593,258 (7.3%)	663,347 (6.1%)	663,915 (7.3%)	Regime I (Physics Opt.)
0	None	961,024 (3.2%)	824,393 (5.4%)	755,794 (5.8%)	756,257 (6.4%)	Regime II (Repr. Adv.)
10	All	753,606 (11.2%) [†]	742,211 (7.4%)	786,366 (6.7%)	789,094 (7.6%)	Regime I (Recal. Suff.)
10	None	1,008,743 (4.8%)	928,028 (6.2%)	866,753 (6.1%)	874,156 (6.6%)	Regime II (Repr. Adv.)
20	All	896,198 (14.0%) [†]	841,734 (7.1%)	875,838 (7.4%)	879,750 (8.0%)	Regime I (Recal. Suff.)
20	None	1,062,646 (6.1%)	1,010,530 (6.2%)	947,869 (6.7%)	959,934 (7.1%)	Regime II (Repr. Adv.)
30	All	1,069,266 (16.6%) [†]	939,190 (7.5%)	987,760 (7.1%)	990,091 (7.6%)	Regime I (Recal. Suff.)
30	None	1,138,774 (7.5%)	1,101,292 (6.7%)	1,054,592 (6.4%)	1,064,238 (6.7%)	Regime II (Repr. Adv.)
60	All	1,656,285 (24.0%) [†]	1,228,609 (9.5%)	1,283,855 (9.6%)	1,274,047 (9.6%)	Regime III (Tail Risk)
60	None	1,437,280 (11.8%) [†]	1,378,900 (9.0%)	1,330,318 (8.7%)	1,331,840 (8.8%)	Regime II (Repr. Adv.)

[†]Exceeds nominal 10% violation target ($q^* = 0.90$). Bold indicates lowest compliant surrogate screening metric. Under $\tau = 60$ min, by-seed compliance is 3/5 seeds (60%) across models.

In contrast, learned representations (STGQ-Modular) reconstruct operating states from secondary electromechanical and spatio-temporal consequences: 1. *Active power divergence:* Rotor aerodynamic torque shedding is inferred from the relationship between active power P_{atv} and inflow velocity; 2. *Electrical transients:* Reactive power fluctuations and voltage dynamics signal pitch control actions; 3. *Wake context:* Upstream operating states propagate downstream through dynamic wake advection.

TABLE II

SEED-PAIRED OPERATIONAL HYPOTHESIS TESTS AND MECHANISM ABLATION CONTRASTS ($h = 6$, 1-HOUR AHEAD DISPATCH, COST RATIO $\rho = 10$).

Regime	Evaluated Contrast / Model	APSREI (kW · h)	95% Bootstrap CI	Operational Verdict & Hypothesis Test
Clean	Cont. Phys. vs. STGQ-Routed			
	(Physical Dominance)	-77,659	[-106,175, -49,143]	Physical rule strictly superior ($p < 0.01$; 95% CI excludes zero)
	STGQ-Dense vs. STGQ-Routed			
	(MoE Mechanism Ablation)	+32,154	[-1,148, +65,456]	Routing advantage falsified (95% CI includes zero, $p = 0.380$)
	Global Quantile vs. STGQ-Routed	+39,962	[+12,556, +67,367]	Regime-blind baseline penalty (95% CI excludes zero)
	Missingness GBDT vs. STGQ-Routed	+279,576	[+249,539, +309,613]	Significant tabular model deficit (95% CI excludes zero)
Delay-6	Frozen Direct MLP vs. STGQ-Routed	+690,695	[+448,769, +932,621]	Severe tail conservatism in uncalibrated direct MLP
	Cont. Physical Quantile			
	(Boundary Breakdown)	-27,126	[-46,907, -7,346]	Unadapted physical rule breaches target (12.20% violation)
	STGQ-Dense vs. STGQ-Routed			
	(MoE Mechanism Ablation)	+16,267	[+7,131, +25,402]	Minor 1.40% marginal difference; dense baseline sufficient
	Global Quantile vs. STGQ-Routed	+17,010	[+3,486, +30,535]	Boundary representation recovery (95% CI excludes zero)
Noise	Missingness GBDT vs. STGQ-Routed	+156,411	[+140,004, +172,817]	Significant tabular model deficit under latency
	Frozen Direct MLP vs. STGQ-Routed	+518,658	[+307,212, +730,104]	Severe tail conservatism in uncalibrated direct MLP
	Cont. Phys. vs. STGQ-Routed			
	(Physical Dominance)	-28,804	[-47,555, -10,053]	Physical rule holds under zero-delay sensor noise
	STGQ-Dense vs. STGQ-Routed			
	(MoE Mechanism Ablation)	+21,006	[+2,419, +39,594]	Minor 2.05% difference between dense and routed heads
Markov	Global Quantile vs. STGQ-Routed	+26,475	[+9,121, +43,828]	Baseline penalty under noise
	Missingness GBDT vs. STGQ-Routed	+287,563	[+248,847, +326,279]	Significant tabular model deficit
	Frozen Direct MLP vs. STGQ-Routed	+612,832	[+413,733, +811,931]	Severe tail conservatism in uncalibrated direct MLP
	Cont. Phys. vs. STGQ-Routed			
	(Physical Dominance)	-82,754	[-111,026, -54,482]	Physical rule superior during packet drop if unlagged
	STGQ-Dense vs. STGQ-Routed			
	(MoE Mechanism Ablation)	+29,453	[-2,874, +61,779]	Routing advantage falsified (95% CI includes zero)
Markov	Global Quantile vs. STGQ-Routed	+39,506	[+15,961, +63,052]	Baseline penalty under burst packet drop
	Missingness GBDT vs. STGQ-Routed	+271,316	[+245,044, +297,588]	Significant tabular model deficit
	Frozen Direct MLP vs. STGQ-Routed	+689,714	[+448,689, +930,739]	Severe tail conservatism in uncalibrated direct MLP
	Frozen Direct MLP vs. STGQ-Routed			

Note: Statistical verdicts are based on empirical 95% confidence intervals computed via day-block cluster bootstrap over

35 test days (1,000 resamples of complete diurnal blocks). STGQ-Routed serves strictly as an architectural mechanism-ablation comparator testing dynamic MoE gating against unrouted representations. Inconclusive difference denotes 95% CI crossing zero.

As detailed in Table~III, STGQ-Modular achieves 755,794 kW · h at $\tau = 0$ (an 8.3% saving of 68,599 kW · h over recalibrated physics) and maintains consistent reserve screening reductions of 46.7k–68.6k kW · h across all latencies. Furthermore, learned representations double transition-window recall (0.416 vs. 0.196 for degraded physics). This confirms that representation learning provides essential utility when primary aerodynamic state variables are withheld.

E. Result 5: Classification Accuracy Does Not Equal Reserve Value (RQ3)

A critical question in operational machine learning is whether superior classification accuracy translates directly into superior reserve screening performance. To interrogate this hypothesis, we benchmark an independent sequence classifier (Graph WaveNet coupled to a live-anchor classifier) against joint representation models using empirical records from completed control benchmarks (modular_classifier_reserve_summary.csv).

Under nominal telemetry, the independent classifier achieves perfect transition recall (1.000) and high precision (0.879), surpassing the joint boundary router (0.960 recall, 0.758 precision). However, when evaluated on pre-dispatch reserve screening, the independent classifier incurs a surrogate reserve cost of 86,536,321 kW · h (9.54% violation). It significantly trails both the joint boundary-forced router (84,577,217 kW · h, 9.00% violation) and the physical-bin baseline (84,314,353 kW · h, 9.31% violation) by over 2.22M kW · h ($p < 0.01$).

This operational penalty occurs because discrete classification errors in independent models are decoupled from forecast residual magnitudes. Independent classifiers frequently misclassify regimes during steep wind-speed ramps where forecast residuals are largest, incurring disproportionately severe Newsvendor penalties. In contrast, joint representation models couple latent state awareness directly with spatio-temporal feature embeddings, ensuring that regime uncertainty is reflected in residual quantile sizing. This proves that classification accuracy is not equivalent to reserve screening value.

F. Result 6: MoE Mechanism Falsification (RQ3)

Benchmarking capacity-matched neural architectures directly tests whether dynamic Mixture-of-Experts routing provides operational benefits over unrouted dense representations. Across all evaluated conditions, the hypothesis that dynamic expert routing is the source of reserve screening improvements is **falsified**:

1) Clean Telemetry: At $h = 1$, STGQ-Dense and STGQ-Routed achieve statistical parity (690,614 vs. 660,990 kW · h, ratio 1.045; seed-paired difference +29,624 kW · h, 95% CI $[-53,774, +113,023]$, $p = 0.380$). At $h = 6$, the paired difference +32,154 kW · h likewise crosses zero ($[-1,148, +65,456]$, Table~II).

2) Multi-Horizon Seed Analysis: In multi-horizon dispatch reserve-screening records across seeds 201–205 (dense_pricing_by_seed.csv), the 5-seed mean surrogate cost for soft-gate-binning is 16.065M kW · h compared to 16.076M kW · h for soft-dense-binning (a difference of <

0.07%). In fact, dense classification achieves lower surrogate cost than gated MoE in three out of five seeds (Seeds 203, 204, and 205).

3) Decoupled Modular Architecture: STGQ-Modular (a decoupled architecture pairing frozen spatio-temporal representations with a dedicated residual quantile head) achieves $1,284,098 \pm 114,629$ kW · h and $9.7\% \pm 1.1\%$ violation under Delay-6 at $h = 1$ (Table~III), matching or outperforming STGQ-Routed ($1,412,321$ kW · h, $10.6\% \pm 1.1\%$).

Point-Forecast Price of Operating-Regime Awareness: In point forecasting, the boundary-aware model achieves an RMSE of 236.13 ± 8.41 , compared with 224.34 ± 2.23 for iTransformer and 225.74 ± 2.60 for Graph WaveNet. This architecture is **deliberately non-RMSE-optimal**: it trades a +2.5% whole-sample error margin to embed operating-regime awareness and prevent catastrophic tail shortfalls. Unconstrained models trigger 78.3–82.5 MWh of unhedged exposure, whereas boundary-aware representations restrict exposure to 53.9 MWh (a 31%–35% reduction).

These findings confirm that the operational benefit stems from consequence-based latent state representation and calibrated residual quantile estimation, not from dynamic expert specialization.

G. Result 7: Deployment, Abstention, and Cross-Farm Boundaries

Failure of Heuristic Selective Abstention: Benchmarking selective decision abstention against random rejection and uniform reserve margin expansion (Supplementary Table~A10f) reveals that heuristic uncertainty scores ($U = \Delta r + 50H(\pi)$) fail under multi-step latency: selective refusal at confidence threshold $c = 0.90$ yields a fleet violation rate of 9.44%, which is statistically indistinguishable from random rejection (9.50%, $p > 0.40$). In contrast, uniform reserve margin expansion achieves an 8.29% violation rate while saving 37,823 kW · h, strictly Pareto-dominating heuristic refusal. Stale telemetry degrades both point forecasts and uncertainty estimators, proving that generic predictive uncertainty cannot identify unsafe operating states under prolonged telemetry degradation.

Cross-Farm Scalability and Local Retraining: Table~IV benchmarks pre-dispatch reserve screening performance across four commercial wind farms under local retraining.

TABLE IV
CROSS-FARM PRE-DISPATCH RESERVE SCREENING PERFORMANCE ACROSS FOUR
COMMERCIAL WIND FARMS UNDER LOCAL RETRAINING (5 SEEDS 201–205,
 $\rho = 10$).

Farm	Turbines / Phys. Model	Phys. Recal. (kW·h)	Global Quantile	STGQ Local Retrain	Δ PSREI vs. Global [95% CI]	Operational Finding
WTB	134 / GW	841,734	893,462	855,462	–38k [–67k, –12k]	Supported reserve reduction ($p < 0.05$)
	4 / MM82	185,100	188,400	231,400	+43k [–28.7k, +141.2k]	
LHB	6 / MM92	1,025,000	1,065,000	1,025,000	–40k [–204k, +172k]	Overfitting boundary (CI includes zero)
Kelmarsh	14 / MM82	2,029,000	4,167,000	2,029,000	–2.14M [–3.25M, –1.36M]	Neutral (CI crosses zero; recal. suffices)
Penmanshiel	14 / MM82	2,029,000	4,167,000	2,029,000	–2.14M [–3.25M, –1.36M]	Commercial benefit ($p < 0.05$; 2.14M saved)

In utility-scale arrays (WTB, 134 turbines; Penmanshiel, 14 turbines), spatial wake modeling delivers substantial reserve screening improvements (Δ PSREI = –38k and –2.14M kW · h, both $p < 0.05$). Conversely, on micro-farms (Kelmarsh, 6

turbines; LHB, 4 turbines), spatial wake redundancy is absent: Kelmarsh yields a neutral bootstrap difference ($-40\text{k kW} \cdot \text{h}$, 95% CI crossing zero), while LHB exposes an **overfitting boundary** where graph convolutions overfit localized terrain features ($+43\text{k kW} \cdot \text{h}$ penalty, $[-28.7\text{k}, +141.2\text{k}]$). Furthermore, zero-shot cross-farm transfer exhibits pronounced directional asymmetry (Kelmarsh $\rightarrow$ Penmanshiel NMI 0.77 vs. Penmanshiel $\rightarrow$ Kelmarsh NMI 0.34). Reliable industrial deployment strictly mandates site-specific local retraining.

Point of Common Coupling (PCC) Portfolio Smoothing:

Aggregating turbine power flows at the PCC bus ($P_{\text{farm}} = \sum_{i \in \mathcal{T}_t} P_{i,t}$, ~ 121 turbines) unlocks spatial diversification: turbulence cancels across the array, leaving boundary transitions as primary reserve risk. Diversification concentrates in transitional regimes (10%–90% pitching), saving $-1.33\text{M kW} \cdot \text{h}$ (CI $[-1.68\text{M}, -1.07\text{M}]$) over continuous physical pitch. Decadal records (Kelmarsh, 9 years; Penmanshiel, 8.6 years) confirm operational recalibration under climatological drift.

VI. DISCUSSION: THE PRINCIPLE OF MINIMUM SUFFICIENT MODEL COMPLEXITY

The empirical findings dismantle the conventional dichotomy between pure physical modeling and unconstrained deep learning, establishing an operational principle of **Minimum Sufficient Model Complexity** governed by channel observability and telemetry freshness:

1) *What Information Does Physics Require:* Deterministic aerodynamic curves evaluate pristine telemetry ($\tau = 0$, full observability) with strict asymptotic power conservation ($\partial P / \partial v \propto v^2$ below rated speed, $P \equiv P_{\text{rated}}$ above rated speed). Avoiding out-of-distribution neural variance, they achieve optimal surrogate reserve screening ($589.5\text{k} - 881.4\text{k kW} \cdot \text{h}$, compliant $\sim 6.8\%$ violation; Tables~I and~III). Under fresh, observable aerodynamics, deep representation learning is methodologically superfluous and operationally sub-optimal.

2) *When Does Recalibration Break Down:* Under observable transmission latency (10–30 min), stale tuples ($v_{t-\tau}, \beta_{t-\tau}$) induce misclassification, but state-conditional recalibration absorbs 55.3% of shortage loss ($127.6 \rightarrow 57.0$ MWh) and maintains compliant violation rates (7.1%–7.5%) without updating neural weights. Offline recalibration breaks down only under contingency horizons ($\tau = 60$ min, Table~A111, 13.8% violation), where extrapolation across the cubic sensitivity cliff exceeds static buffer margins.

3) *What Learned Representations Actually Know:* Decisive utility emerges when blade-pitch telemetry is withheld (no_pitch), where physical rules inflate penalties to $1,378,900\text{ kW} \cdot \text{h}$. The spatio-temporal encoder reconstructs latent regimes from secondary SCADA consequences: (i) active power divergence reflecting aerodynamic torque shedding; (ii) electrical and reactive transients signaling control action; and (iii) spatial wake advection providing topological corroboration. Counterfactual probes confirm substantial mutual information (0.302–0.674 NMI vs. $< 10^{-4}$ for chance controls), reducing penalties by $46.7\text{k} - 68.6\text{k kW} \cdot \text{h}$.

4) *Decision Relevance of the Boundary Posterior:* Regime

classification accuracy is decoupled from forecast residual magnitude: an independent classifier achieves high transition recall (1.000) yet incurs a severe penalty ($86.54\text{M kW} \cdot \text{h}$ vs. 84.31M for physics) by misclassifying steep ramps where residuals peak. Reserve screening requires joint representation and residual quantile coupling. Crucially, dynamic Mixture-of-Experts routing provides no measurable advantage over unrouted dense baselines (cost ratio 1.045, $p = 0.380$; seed difference $< 0.07\%$), while decoupled modular regression (STGQ-Modular) achieves equivalent tail reliability (9.7% violation).

5) *Industrial Deployment Boundaries and Scalability:* System operators must observe three firm deployment guardrails: (i) *Array scale sensitivity:* utility-scale arrays (WTB, Penmanshiel) yield substantial savings (-38k to $-2.14\text{M kW} \cdot \text{h}$, $p < 0.05$); compact micro-farms (LHB, 4 turbines) lack wake redundancy, causing graph convolutions to overfit local terrain ($+43\text{k kW} \cdot \text{h}$ penalty); (ii) *Local retraining mandate:* directional transfer asymmetry (Kelmarsh $\rightarrow$ Penmanshiel NMI 0.77 vs. reverse 0.34) requires site-specific retraining; and (iii) *Abstention realism:* heuristic selective abstention fails under latency (9.44% vs. 9.50%, $p > 0.40$). Operators should deploy uniform reserve margin expansion during severe telemetry outages.

VII. LIMITATIONS

To provide industrial practitioners with transparent boundaries of applicability, we delineate four foundational limitations of this study:

1) *Synthetic Stress versus Industrial Telemetry Contingencies:* Telemetry impairments in primary benchmarks are modeled via controlled synthetic processes: 10–30 min delays represent buffer backlogs, while 60-min latency and Gilbert-Elliott Markov dropouts evaluate severe contingency stress bounds. Ground-truth operating regimes on WTB rely on anemometer wind speed and collective pitch pseudo-labels. While withheld-channel tests confirm aerodynamic regimes are recoverable across four commercial plants, field turbines may experience non-stationary instrumentation drift, sensor icing, or uncoordinated individual pitch control actions not captured in supervisory SCADA feeds.

2) *Risk Metric Surrogacy and Unmodeled Grid Dynamics:* Evaluation operates strictly at Level-1 pre-dispatch operating reserve screening. The penalized reserve shortfall index (PSREI at $\rho = 10$) serves as a localized newsvendor risk surrogate. This abstraction intentionally omits Level-2 bulk transmission power-flow physics (full AC-OPF constraints, thermal congestion, voltage limits), dynamic unit commitment of thermal reserves, Locational Marginal Prices, and two-settlement wholesale balancing cashflows. Realized operational balancing savings depend on specific system-wide market structures and penalty tariffs.

3) *Horizon, Sample Size, and Aerodynamic Calibration Tolerances:* Empirical benchmarks are bounded to a 35-day continuous test split across five neural random seeds (201–205). While capturing diurnal transitions and winter storm

fronts, this window does not encompass full multi-annual climatological cycles or seasonal air-density shifts. Moreover, aerodynamic parameters exhibit OEM-specific calibration tolerances: rated wind speeds range from 12.5 m/s (Senvion MM92) to 14.5 m/s (Senvion MM82) and cut-in transitions differ across turbine controllers, requiring OEM-specific threshold calibration.

4) Field Deployment Overhead and Cross-Farm Overfitting Boundaries: Real-time substation edge deployment requires balancing neural inference latency against supervisory RTU polling cycles. While the modular residual head requires minimal computational overhead, spatial graph convolutions overfit localized topography on small arrays (LHB, 4 turbines, +43k kW · h penalty). Zero-shot cross-farm transfer exhibits pronounced directional asymmetry (pooled NMI 0.557; Tables~A8–A9d), establishing that reliable deployment strictly demands site-specific local retraining rather than uncalibrated cross-site transfer.

VIII. CONCLUSION

This study fundamentally repositions machine learning in wind plant operating reserve screening from a presumed universal replacement for physical models to a targeted, conditional inference mechanism under telemetry degradation. Across five-seed evaluations on the 134-turbine WTB plant and multi-year data from three European facilities, we establish three primary conclusions:

First, under fresh and fully observable SCADA telemetry, deterministic aerodynamic power curves minimize surrogate reserve screening penalties (589,535 kW · h at $h = 1$, 881,367 kW · h at $h = 6$, ~ 6.8% violation), outperforming deep neural networks. Deterministic rules experience catastrophic reliability breakdown only when telemetry staleness forces them to extrapolate across the cubic-to-flat sensitivity cliff near rated wind speed, surging violations to 24.0%.

Second, when telemetry channels remain observable under transmission delays, simple state-conditional recalibration absorbs 55.3% of shortage exposure (127.6 → 57.0 MWh) without neural representation learning. Learned representations acquire demonstrable utility only when blade-pitch registers become unobservable: by inferring latent operating states from secondary electromechanical and spatio-temporal consequence channels, learned representations outperform recalibrated physical rules by 46.7k–68.6k kW · h across all latencies and double transition recall (0.416 vs. 0.196).

Third, dynamic Mixture-of-Experts routing confers no statistically detectable advantage over unrouted dense or modular architectures ($p = 0.380$), while decoupled modular residual regression matches tail reliability (9.7% violation). Furthermore, heuristic selective decision abstention is Pareto-dominated by uniform reserve margin expansion under persistent latency.

Consequently, modern wind plant dispatch systems should implement an observability-aware division of labor: deploying deterministic aerodynamic rules under clean telemetry, lightweight recalibration under observable drift, and reserving

learned latent-state representations for regimes where critical physical states are unobservable.

IX. AI USE STATEMENT

The authors used OpenAI ChatGPT/Codex only for language editing, consistency checks, and submission-material drafting; all data, analyses, references, conclusions, and final text were reviewed and controlled by the authors.

X. CODE AND DATA AVAILABILITY

The raw KDD Cup 2022, ENGIE La Haute Borne, Kelmarsh, and Penmanshiel SCADA datasets are public (raw third-party data not redistributed). Code, configurations, releasable derived tables, figure data, and checkpoints will be made available with the article. The reproduction package covers WT-B routing, reserve audit, anchor-stress caches, early-warning degradation, classifier control, class-weight sensitivity, and evidence-freeze protocols across declared seeds. The analysis involves no human subjects.

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